Master Thesis Abstract

Modeling and Predicting Daily Dengue Cases in Bangladesh Using Meteorological Factors: A Comparative Study of Statistical and Machine-Learning Approaches

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Abstract

Background and objectives. Dengue is a major climate-sensitive public-health problem in Bangladesh, but explaining how weather relates to reported cases and forecasting short-term burden are distinct analytical tasks. This thesis examined meteorological associations with daily reported dengue counts and compared seven-day-ahead forecasting approaches across all eight administrative divisions.

Data and methods. A retrospective, two-track ecological spatio-temporal design was used. Track A linked daily dengue counts from the Directorate General of Health Services (DGHS) with temperature, rainfall, relative humidity and sunshine duration from the Bangladesh Meteorological Department (BMD), 2022–2024 (8,768 division-days). Poisson and Negative Binomial regression were compared, with Negative Binomial type 2 (NB2) as the primary model, a prespecified lag comparison, a nonlinear temperature term, and sensitivity analyses for serial dependence (AR(1) GEE; PPML) and excess zeros (zero-inflated NB). Track B used a separate leakage-controlled dataset (10,904 origin–target records) limited to information available at forecast origin, with weather from a distinct reanalysis-based series. Persistence, Log-Ridge, Random Forest and XGBoost were compared by rolling-origin validation and a final temporal test (January–October 2025), with SHAP and feature ablation for interpretation.

Results. Of 483,715 cases reported in 2022–2024, 320,644 occurred in 2023. Counts were severely overdispersed (Poisson Pearson dispersion ≈30), favouring NB2. A 28-day lag minimised AIC, though 14 days fitted almost as well. Temperature showed the most robust nonlinear, delayed adjusted association; rainfall and sunshine showed no robust independent linear association; humidity was dependence-sensitive. In the 2025 test, Log-Ridge had the lowest division-day mean absolute error (MAE 8.854), ahead of persistence (9.451), XGBoost (9.792) and Random Forest (11.144). Recent dengue history dominated predictive information: adding history to weather and structural features reduced development MAE by 71.47%, whereas adding meteorology to history and structural features reduced it by 8.22%. Error rose sharply at epidemic peaks (MAE 84.86, bias -83.34 at or above the 95th-percentile burden).

Conclusions and implications. Explanatory association and predictive utility are complementary but distinct. Short-horizon forecasting should prioritise timely surveillance history, retain meteorology as supplementary context, and monitor peak-burden error and uncertainty. The findings are ecological, rest on passively reported surveillance counts and three annual cycles dominated by the 2023 epidemic, and do not establish causality; prospective validation is required before operational use. Division-level analysis cannot address urban-scale processes but offers a statistical baseline for finer-scale studies of climate, urban environmental conditions and disease risk, requiring population denominators and land-use, water-body, mobility and entomological data.

Keywords: Dengue; disease surveillance; meteorological factors; Negative Binomial regression; seven-day forecasting; Log-Ridge; XGBoost; SHAP; temporal validation